Sri Lanka Institute of Information Technology (SLIIT)

IT3021 – Data Warehousing and Business Intelligence
Assignment 1

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Degree Program: BSc (Hons) in Information Technology Specializing in Data Science

Submission Date: 29.03.2026

CONTENTS

Task 1 – Dataset Selection 3
1.1 Introduction 3
1.2 Dataset Description 3
1.3 Dataset Attributes 4
1.4 Entity Description 4
Task 2 – Data Sources 5
2.1 Introduction 5
2.2 Data Sources Used 5
2.3 CSV Data Source 5
2.4 SQL Server Data Source 6
Task 3 – DW & BI Solution Architecture 6
3.1 Introduction 6
3.2 DW & BI Solution Architecture Diagram 6
3.3 Architecture Components 7
Task 4 – Data Warehouse Design & Development 8
4.1 Introduction 8
4.2 Dimensional Model 8
4.3 Relational Diagram (Star Schema) 9
4.4 Fact and Dimension Table Description 10
4.5 Slowly Changing Dimension 11
4.6 Implementation in SQL Server 11
4.7 Assumptions 12
TASK 5 – ETL DEVELOPMENT 12
5.1 Introduction 12
5.2 ETL Architecture Overview 13
5.3 ETL Implementation in SSIS 14
5.4 Additional Data Source (Country Risk Levels) 19
5.5 Summary of ETL Process 19
TASK 6 – ACCUMULATING FACT TABLE 22
6.1 Introduction 22
6.2 Extending the Fact Table 22
6.3 Preparing the Update Dataset 22
6.4 Results 23
6.5 SSIS Package for Fact Updates 23
6.8 Conclusion 24

TASK 1 – DATASET SELECTION

1.1 Introduction

For this assignment, a dataset related to financial transaction monitoring was selected. The dataset represents a banking transaction system used for detecting potential money laundering activities. Financial institutions commonly use such systems to monitor transactions and identify suspicious patterns.

The selected dataset is the **Synthetic Transaction Monitoring Dataset for Anti-Money Laundering (AML)** available on Kaggle. This dataset simulates realistic financial transactions between accounts and contains attributes useful for analyzing suspicious financial behavior.

Dataset Source:
https://www.kaggle.com/datasets/berkanoztas/synthetic-transaction-monitoring-dataset-aml/data

Reference Paper:

B.Oztas,D.Cetinkaya,F. Adedoyin, M. Budka, H. Dogan and G. Aksu, "Enhancing Anti-Money Laundering: Development of a Synthetic Transaction Monitoring Dataset," IEEE International Conference on e-Business Engineering (ICEBE), 2023.

1.2 Dataset Description

The dataset contains transaction records representing financial transfers between sender and receiver accounts. Each transaction contains information related to the payment method, currency, bank location, and whether the transaction is suspicious.

The dataset includes approximately **9.5 million transaction records**, which provides sufficient data for performing ETL operations, building a data warehouse, creating OLAP cubes, and generating analytical reports.

The dataset covers transactions over a continuous time period (Date and Timestamp attributes), satisfying the requirement of having approximately one year of data for analysis.

This dataset is suitable for data warehousing as it represents transactional data that can be modeled into fact and dimension tables for analytical processing.

The dataset also includes multiple typologies related to money laundering behavior and normal transactions.

1.3 Dataset Attributes

- The dataset contains the following attributes:

Attribute	Description
Time	Time of the transaction
Date	Date of the transaction
Sender_account	Account ID of the sender
Receiver_account	Account ID of the receiver
Amount	Transaction amount
Payment_currency	Currency used for payment
Received_currency	Currency received
Sender_bank_location	Country of the sender's bank
Receiver_bank_location	Country of the receiver's bank
Payment_type	Method used to perform the transaction (e.g., credit card, debit card, transfer)
Is_laundering	Indicates whether transaction involves laundering
Laundering_type	Category of laundering typology
timestamp	Timestamp of the transaction
is_suspicious	Indicates whether transaction is suspicious

Table 1

1.4Entity Description

Transaction

Represents a financial transaction between two accounts. Each transaction records the amount transferred, currencies involved, payment type, and whether the transaction is suspicious.

Account

Represents a bank account involved in transactions. Each account may act as a sender or receiver in multiple transactions.

Payment Type

Represents the method used to perform the transaction such as credit card, debit card, cheque, cash deposit, ACH transfer, or cross-border transfer.

Currency

Represents the currencies used during the transaction.

BankLocation

Represents the geographical location of the bank where the transaction originates or ends.

Laundering Type

Represents the classification of the transaction typology such as normal transactions or suspicious laundering patterns.

TASK 2 – DATA SOURCES

2.1 Introduction

To develop the data warehouse, multiple data sources were used. These sources provide the necessary data for extraction, transformation, and loading (ETL) processes. The selected sources include structured data stored in CSV files and relational databases.

2.2 Data Sources Used

Source	Type	Description
clean_AML_Dataset.CSV	CSV file	Transaction data (Date, Time, Sender/Receiver accounts, Amount, Currency, Payment Type)
AML_Source Database	SQL Server	Structured tables such as Accounts, Payment Types, Currencies, Locations
CountryRiskLevels.xlsx	Excel	Additional country risk information used for analysis

Table 2

Multiple data source formats were used to simulate a real-world environment where data originates from different systems. This improves the complexity and realism of the ETL process.

2.3 CSV Data Source

The primary data source is the **clean_AML_Dataset.CSV** file obtained from the AML dataset. This file stores raw transaction records in CSV format. It contains attributes such as

transaction time, date, sender account, receiver account, transaction amount, payment type, currencies used, bank locations, and indicators of suspicious or laundering-related activity.

This dataset acts as the main source for the ETL process when populating the data warehouse.

2.4 SQL Server Data Source

A relational database named **AML_Source** was created in Microsoft SQL Server to store structured data extracted from the CSV dataset.

Several reference tables such as accounts, payment types, currencies, and bank locations were generated from the raw dataset. These tables are stored in relational format and are used during the ETL process to populate dimension tables in the data warehouse.

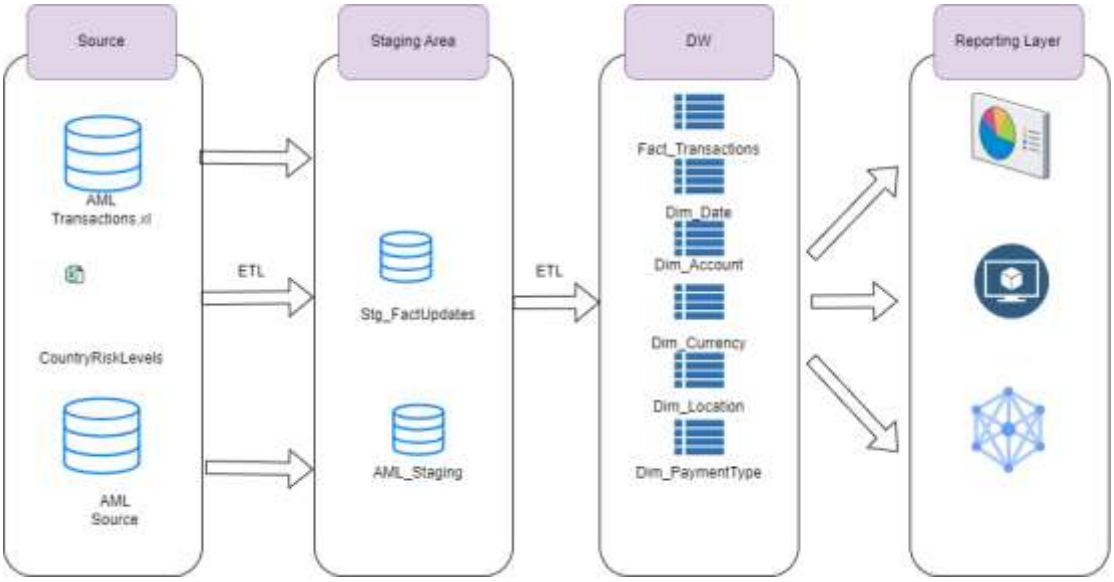
TASK 3 – DW & BI SOLUTION ARCHITECTURE

3.1 Introduction

The Data Warehouse and Business Intelligence (DW & BI) solution architecture describes the overall structure used to collect, transform, store, and analyze data for decision-making purposes.
The proposed architecture integrates data from multiple sources, processes the data through ETL operations, stores the data in a data warehouse, and enables analytical reporting through BI tools.

The architecture is designed to support efficient data integration, analytical processing, and reporting.

3.2 DW & BI Solution Architecture Diagram



3.3 Architecture Components

Data Sources

The system collects data from multiple data sources. In this project, the main data sources include the **clean_AML_Dataset.CSV** dataset and structured tables stored in the AML_Source SQL Server database. These sources contain the raw transaction data and reference data required for analysis.

ETL Process

The ETL (Extract, Transform, Load) process is responsible for preparing the data before loading it into the data warehouse.
During this stage, raw data is extracted from the CSV dataset, cleaned, transformed, and integrated with reference tables from the SQL Server database. The processed data is then loaded into the data warehouse tables.

Data Warehouse

The data warehouse stores integrated and structured data optimized for analytical processing. The warehouse uses a Star Schema design consisting of a central fact table (Fact_Transactions) and several dimension tables such as Dim_Date, Dim_Account, Dim_PaymentType, Dim_Currency, and Dim_Location.

This structure enables efficient analytical queries.

OLAP Cube

An OLAP cube is created using SQL Server Analysis Services (SSAS) to support multidimensional analysis.
The cube allows users to perform operations such as slicing, dicing, and aggregating data across different dimensions such as date, payment type, location, and account.

Reporting and Visualization

The final layer of the architecture provides reporting and visualization capabilities. Business intelligence tools such as Power BI can be used to generate dashboards and reports that help analyze transaction patterns, identify suspicious activities, and support decision making.

TASK 4 – DATA WAREHOUSE DESIGN & DEVELOPMENT

4.1 Introduction

In this task, a data warehouse schema was designed for the AML transaction dataset. The dimensional model was developed to support analytical queries and business intelligence reporting.

A Star Schema design was selected because it simplifies analytical queries and improves query performance. The schema consists of a central fact table representing transaction events and several dimension tables describing contextual attributes of the transactions.

The overall solution follows a three-layer architecture consisting of AML_Source as the raw data source layer, AML_Staging as the intermediate transformation layer, and AML_DW as the final data warehouse layer where the dimensional model is implemented.

4.2 Dimensional Model

The designed dimensional model consists of one fact table and multiple dimension tables.

Fact Table

- Fact_Transactions

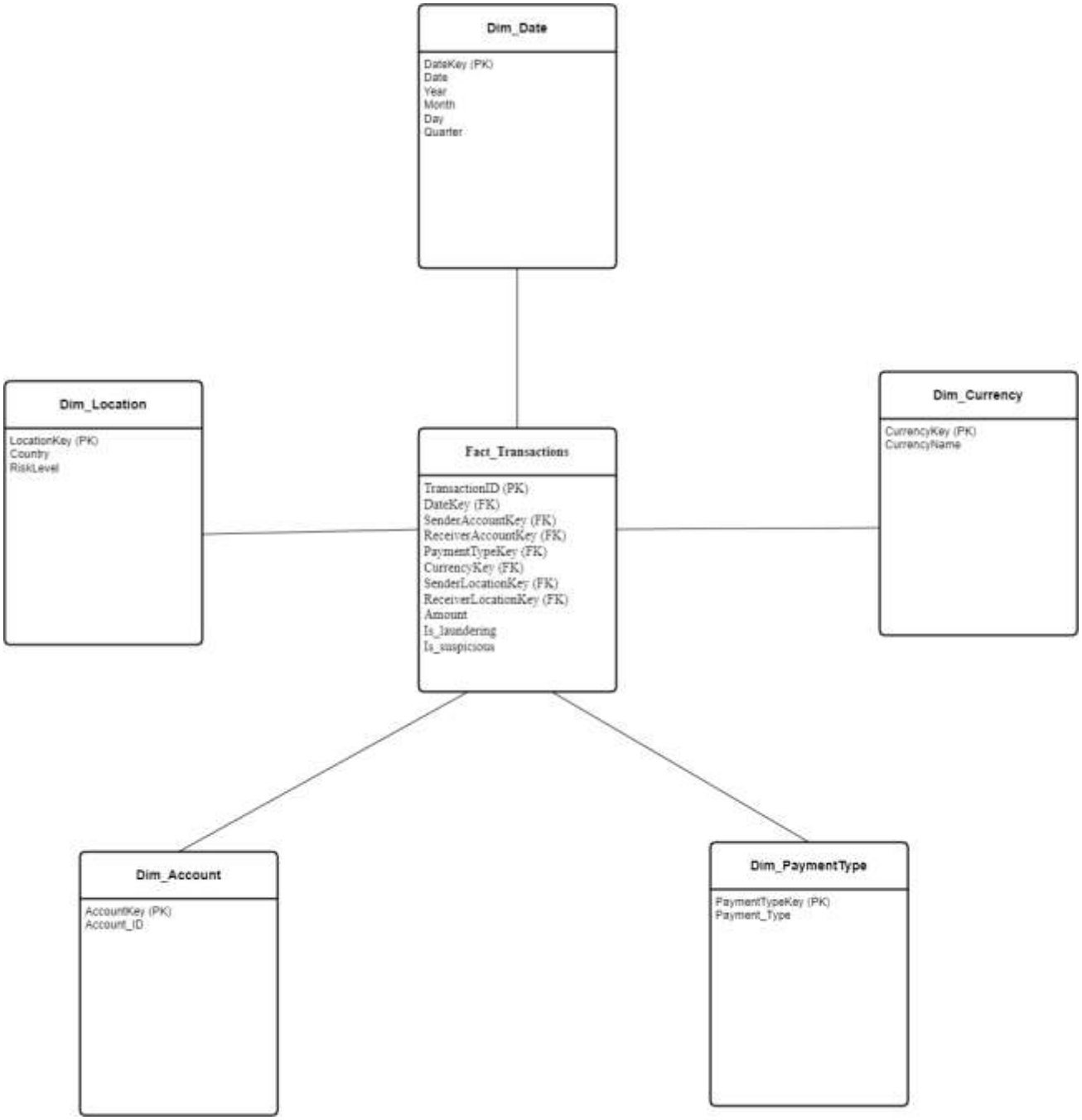
Dimension Tables

- Dim_Date
- Dim_Account
- Dim_PaymentType
- Dim_Currency
- Dim_Location

This design supports analysis of transactions across different dimensions such as time, account, payment method, currency, and bank location.

4.3 Relational Diagram (Star Schema)

The relational diagram of the dimensional model is shown below.



In this schema, the Fact_Transactions table is the center, while all dimension tables connect directly to the fact table.

4.4 Fact and Dimension Table Description

Fact_Transactions

Column	Description
TransactionID	Unique transaction identifier
DateKey	Foreign key to Dim_Date
SenderAccountKey	Foreign key to Dim_Account
ReceiverAccountKey	Foreign key to Dim_Account
PaymentTypeKey	Foreign key to Dim_PaymentType
PaymentCurrencyKey	Foreign key to Dim_Currency
ReceivedCurrencyKey	Foreign key to Dim_Currency
SenderLocationKey	Foreign key to Dim_Location
ReceiverLocationKey	Foreign key to Dim_Location
Amount	Transaction amount
Is_laundering	Indicates laundering activity
Is_suspicious	Indicates suspicious transaction

Dim_Date

Column
DateKey
Date
Day
Month
Quarter
Year

Dim_Account

Column
AccountKey
Account_ID

Dim_PaymentType

Column
PaymentTypeKey
Payment_Type

Dim_Currency

Column
CurrencyKey
CurrencyName

Dim_Location

Column
LocationKey
Country

4.5 Slowly Changing Dimension

The Dim_Location table was implemented as a Slowly Changing Dimension.

This allows the system to track changes in location related information over time while preserving historical data.

Additional fields used for this purpose include:

- StartDate
- EndDate
- IsCurrent

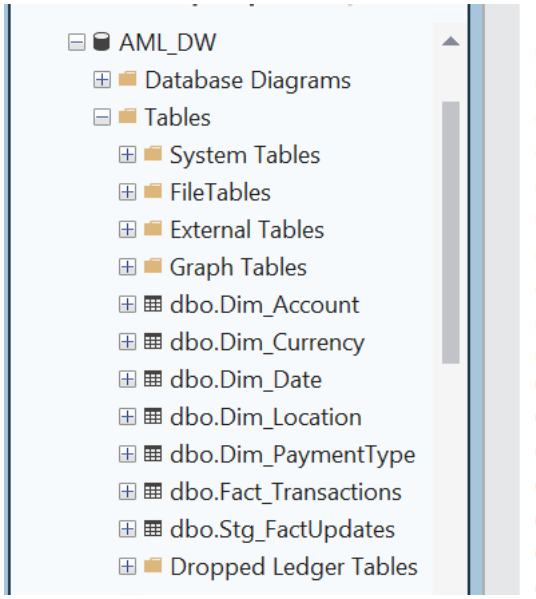
These attributes help maintain the historical versions of records when changes occur.

Type 2 Slowly Changing Dimension methodology was applied to preserve historical changes in location data.

4.6 Implementation in SQL Server

The designed star schema was implemented in Microsoft SQL Server Management Studio (SSMS). SQL scripts were used to create the data warehouse tables including dimension tables and the fact table. Primary keys and foreign key relationships were defined to maintain referential integrity between the tables.

Surrogate keys were used for dimension tables, while business transaction data was stored in the fact table.



4.7 Assumptions

The following assumptions were made during the design of the data warehouse:

1. Each transaction represents a single financial transfer between two accounts.
2. The same Account dimension is used for both sender and receiver accounts.
3. The Date dimension is used to support time-based analysis of transactions.
4. Currency information is stored in a separate dimension to support multi-currency analysis.
5. Bank location information is stored in the Location dimension to support geographical analysis.
6. The star schema was selected to simplify analytical queries and reporting.

TASK 5 – ETL DEVELOPMENT

5.1 Introduction

The ETL (Extract, Transform, Load) process was developed using SQL Server Integration Services (SSIS) to extract data from multiple sources, transform it into a suitable format, and load it into the data warehouse.

The solution follows a three-layer architecture:

- **AML_Source** – Source layer containing raw data
- **AML_Staging** – Intermediate layer for transformation
- **AML_DW** – Final data warehouse layer

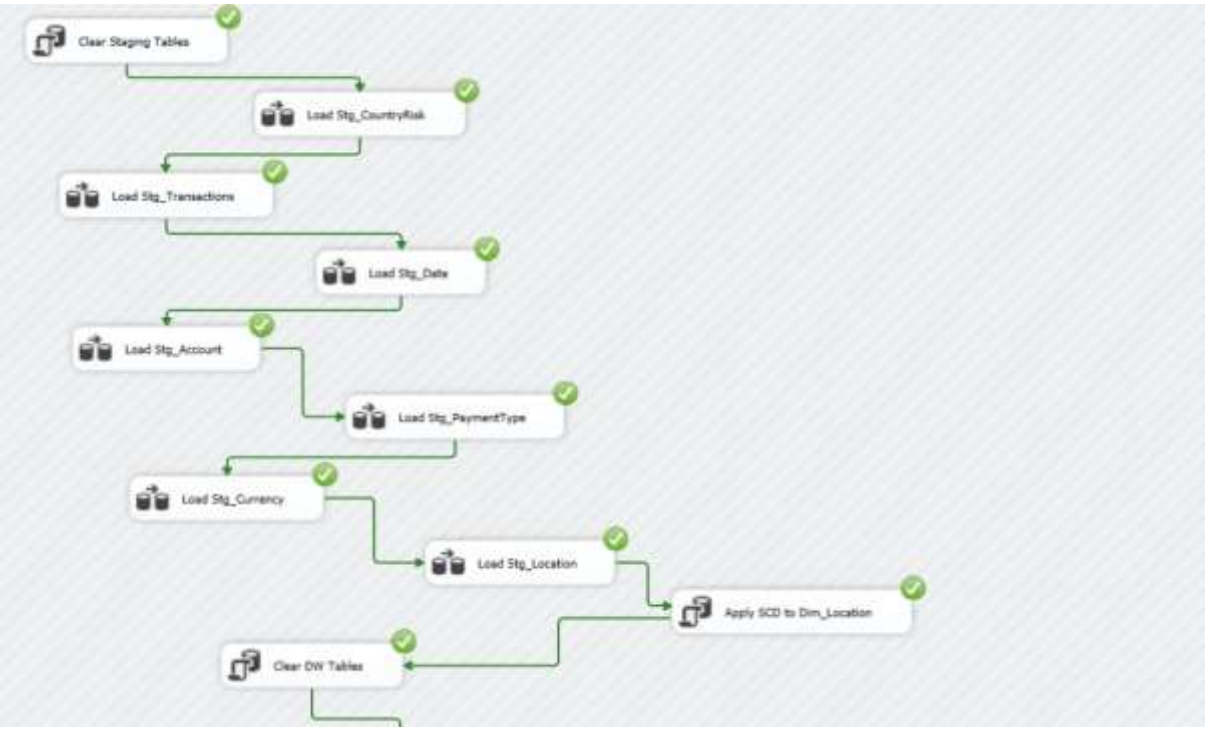
This architecture ensures efficient data processing, data quality, and structured loading into the dimensional model

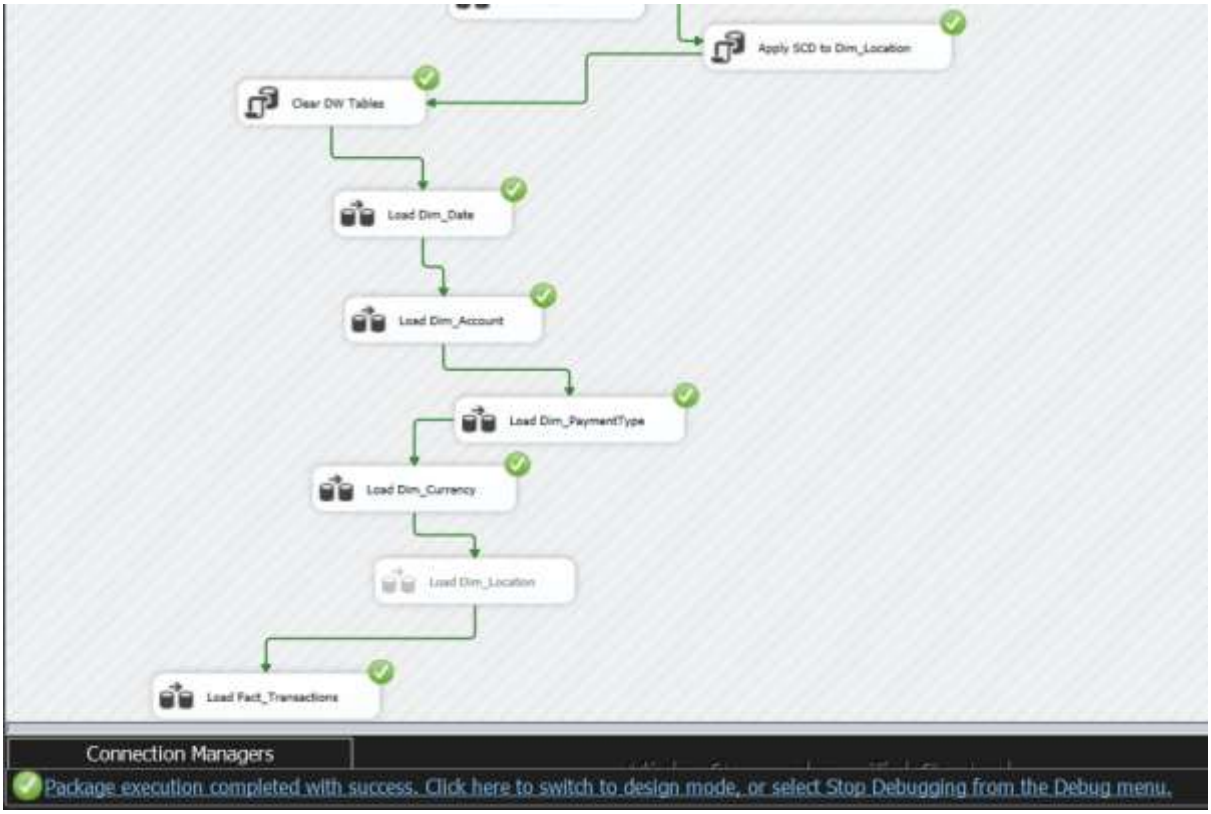
5.2 ETL Architecture Overview

The ETL process follows a structured pipeline:

1. Extract data from AML_Source
2. Load data into AML_Staging tables
3. Perform data transformation and cleansing
4. Load data into dimension tables
5. Perform lookup transformations
6. Load data into Fact_Transactions

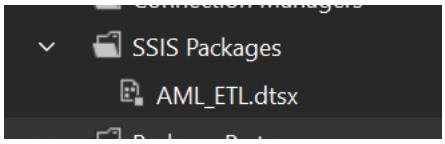
Execution Order:





5.3 ETL Implementation in SSIS

The ETL was implemented using a single SSIS package named:



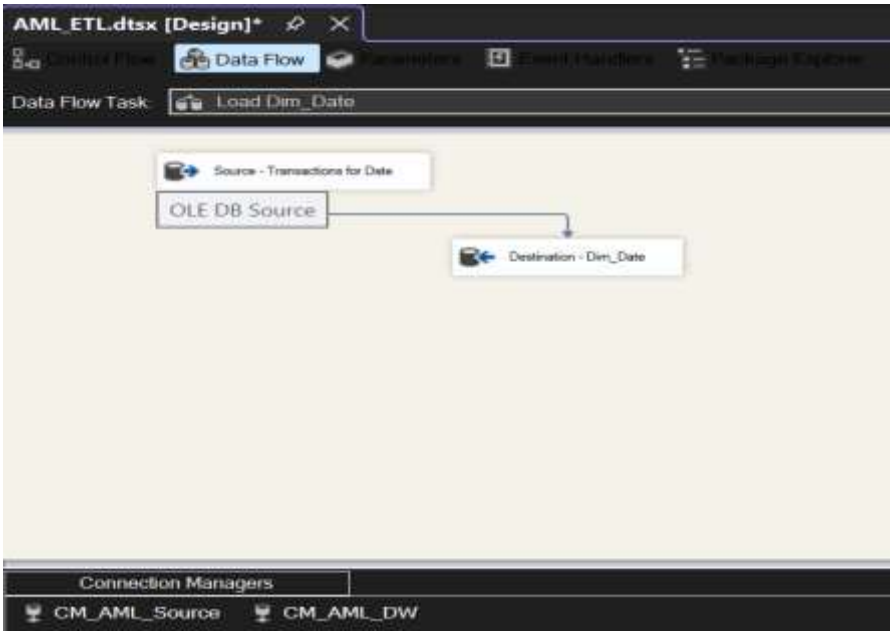
Load Dim_Date

Description

Extracts unique dates from transaction data and generates date attributes.

Transformation

- DateKey generated (YYYYMMDD)
- Extract Day, Month, Quarter, Year



Load Dim_Account

Description

Extracts all unique accounts from sender and receiver.

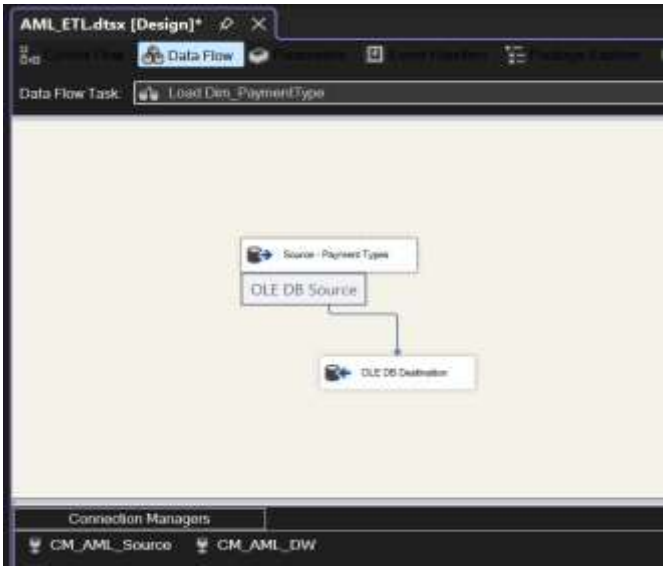
Transformation

- Combine sender and receiver accounts using UNION
- Remove duplicates

Load Dim_PaymentType

Description

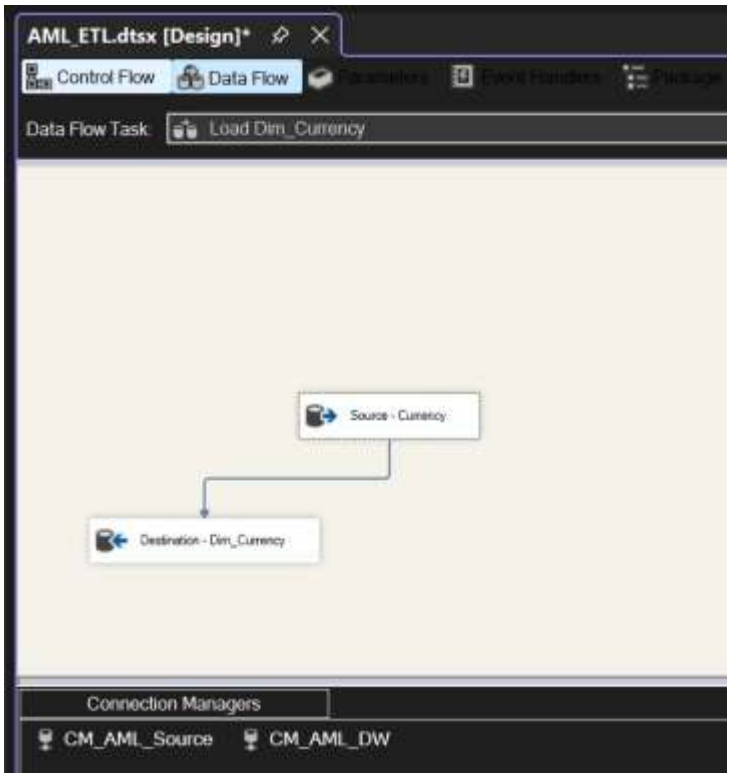
Loads unique payment methods.



Load Dim_Currency

Description:

Loads currencies from both payment and received currency columns.

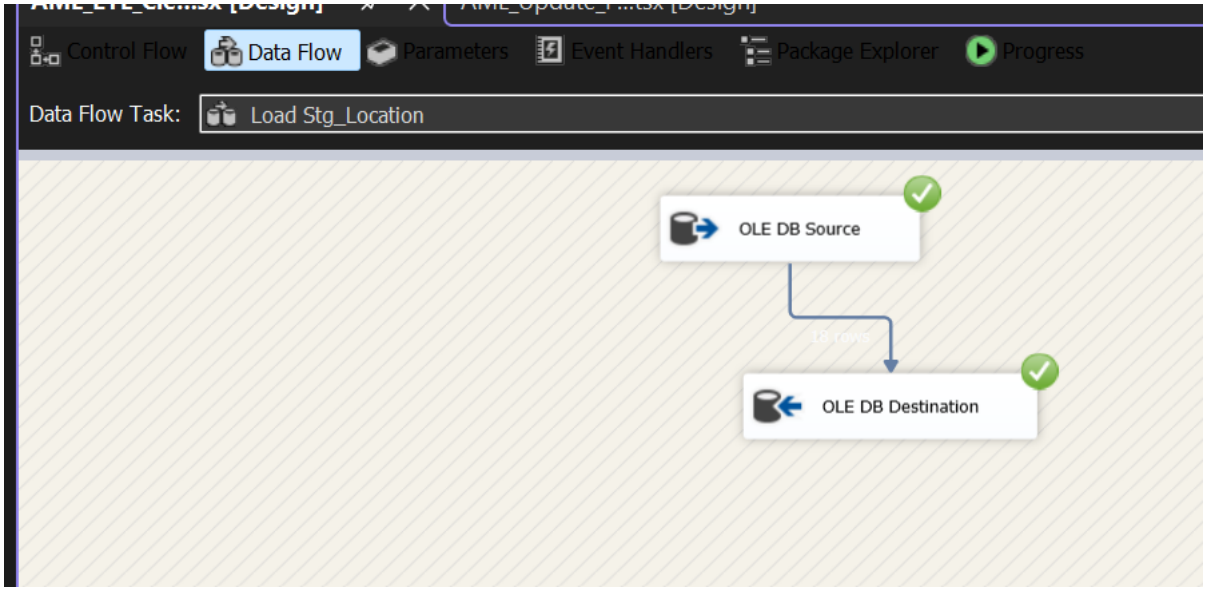


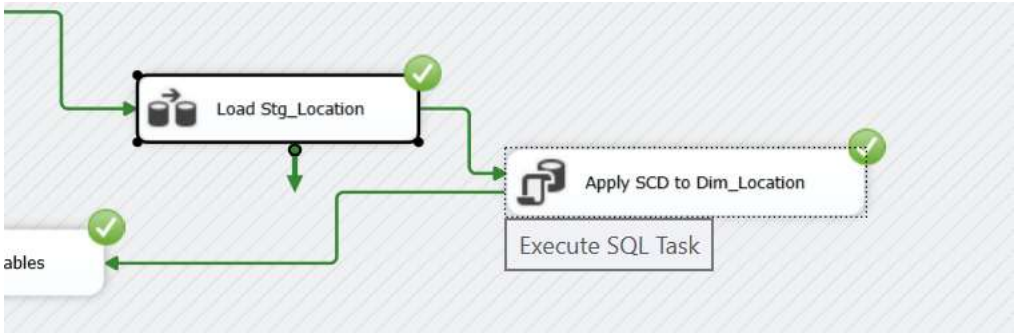
Load Dim_Location

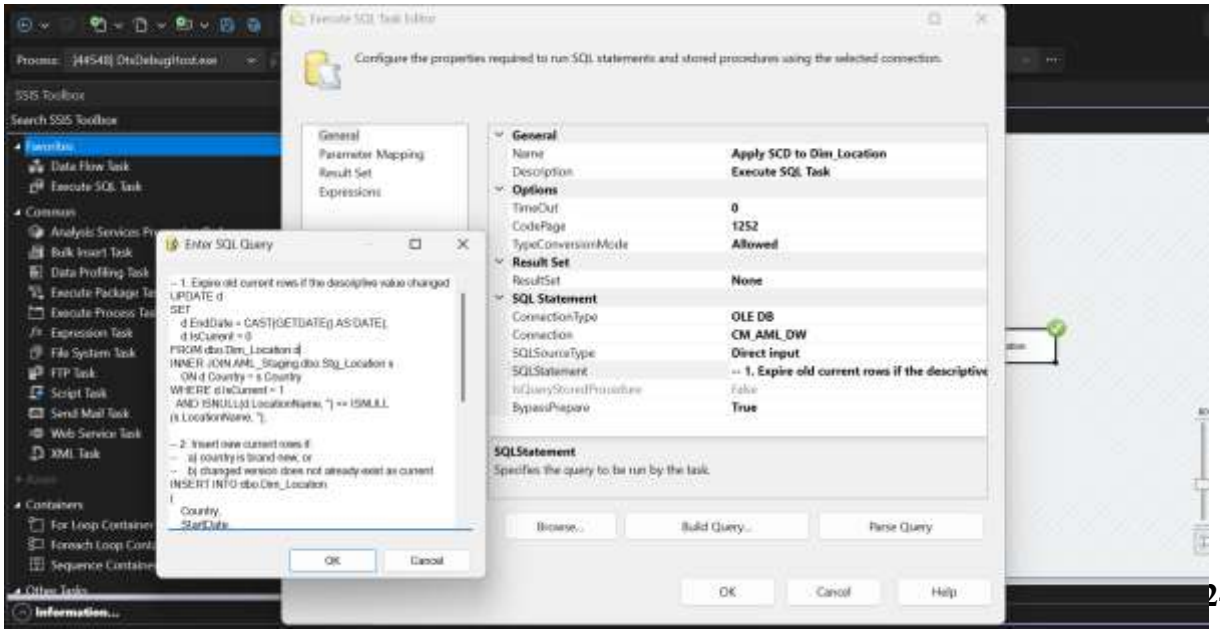
Description

Loads bank locations (countries).

The Dim_Location table was implemented as a Slowly Changing Dimension (Type 2) to preserve historical changes. New records are inserted when changes occur, while existing records are expired using EndDate and tracked using IsCurrent.







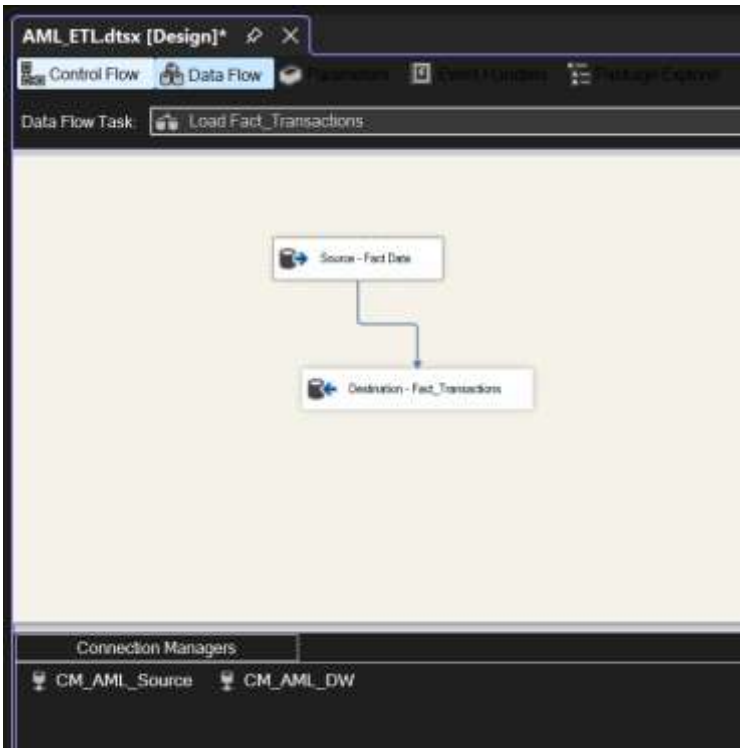
Load Fact_Transactions

Description
This Data Flow Task loads prepared transactional fact data into the Fact_Transactions table in the data warehouse.

Source
The source already contains the required fact attributes and foreign key columns, including DateKey, SenderAccountKey, ReceiverAccountKey, PaymentTypeKey, CurrencyKey, SenderLocationKey, and ReceiverLocationKey.

Transformation
Since the required keys were already available in the source dataset, no additional lookup transformations were required in this Data Flow. The prepared fact data was loaded directly into the destination table.

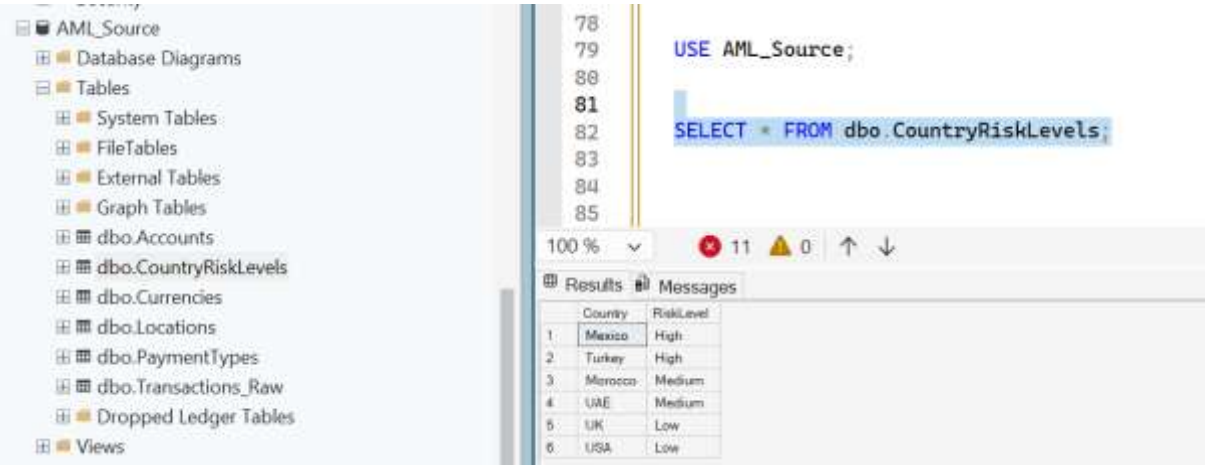
Destination
AML_DW.dbo.Fact_Transactions



5.4 Additional Data Source (Country Risk Levels)

An additional dataset (CountryRiskLevels) was imported from Excel into SQL Server.

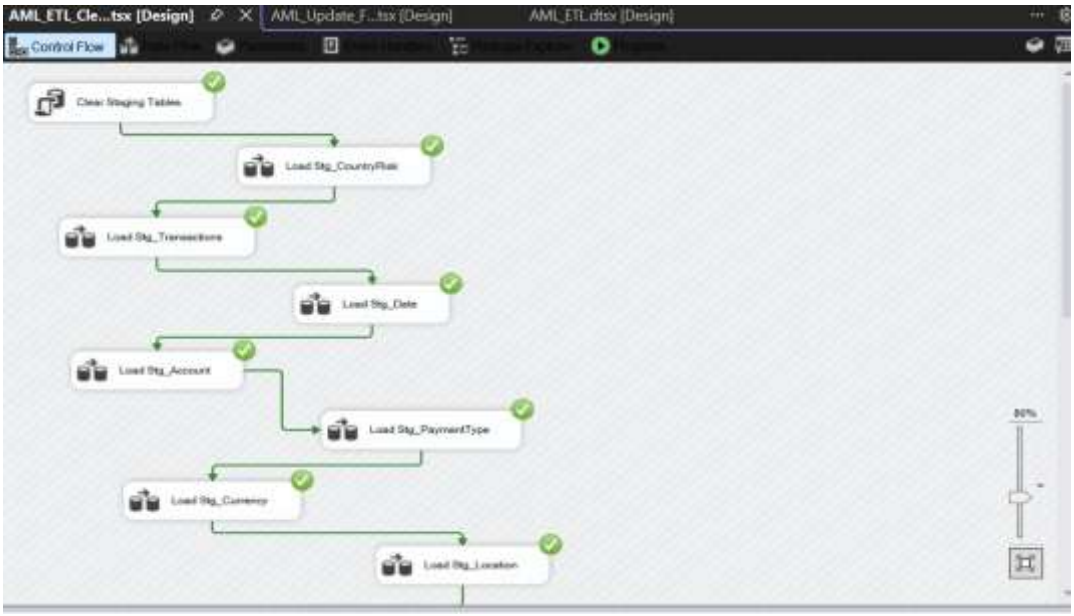
This demonstrates handling multiple data sources.

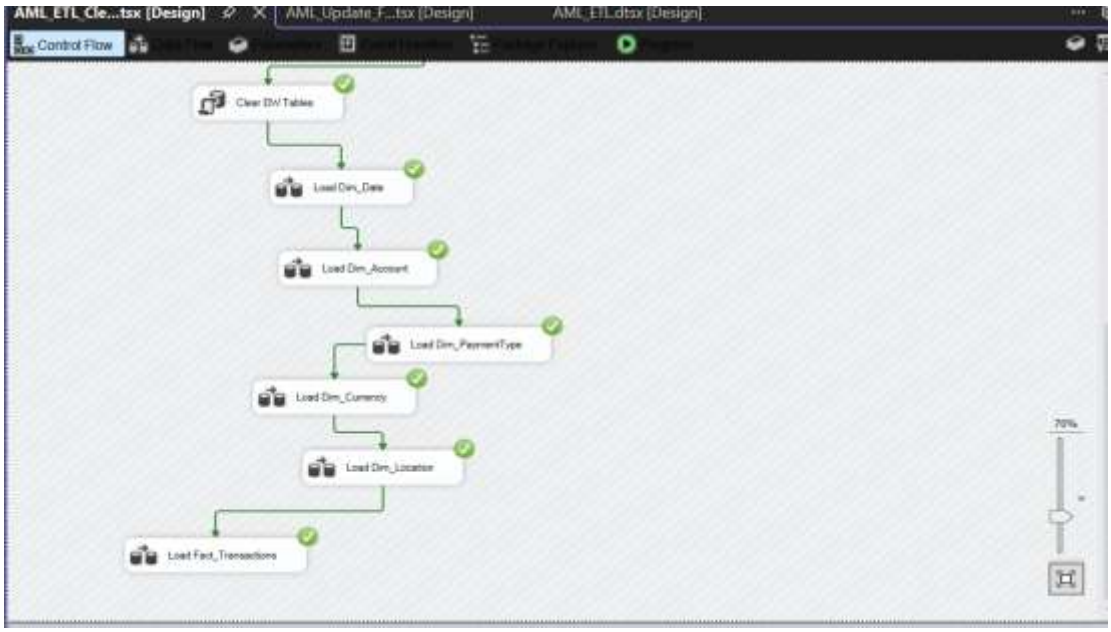


5.5 Summary of ETL Process

- Extracted data from:
  - SQL Server database
  - Excel file
- Applied transformations:
  - Data type conversions
  - Duplicate removal
  - Key generation
  - Data integration via joins
- Loaded into:
  - Dimension tables first
  - Fact table last

This ensures referential integrity and proper star schema population.


AML_ETL_Cle...tsx [Design]
AML_Update_F...tsx [Design]
AML_ETL.dtsx [Design]
Control Flow
Clear Staging Tables
Load Stg_CountryRisk
Load Stg_Transactions
Load Stg_Date
Load Stg_Account
Load Stg_PaymentType
Load Stg_Currency
Load Stg_Location


AML_ETL_Cle...tsx [Design]
AML_Update_F...tsx [Design]
AML_ETL.dtsx [Design]
Control Flow
Clear DW Tables
Load Dim_Date
Load Dim_Account
Load Dim_PaymentType
Load Dim_Currency
Load Dim_Location
Load Fact_Transactions

```sql
USE AML_DW;
GO

SELECT COUNT(*) AS TotalRows FROM Fact_Transactions;
```

100 % No issues found

Results | Messages

	TotalRows
1	503449

```sql
USE AML_DW;
GO

SELECT COUNT(*) AS TotalRows FROM Fact_Transactions;

SELECT TOP 10 * FROM Fact_Transactions;
GO
```

100 % No issues found Ln: 156, Ch: 1 (43 chars, 2 lines) SPC CRLF Windows 1252

Results | Messages

	TransactionID	DateKey	SenderAccountKey	ReceiverAccountKey	PaymentTypeKey	CurrencyKey	SenderLocationKey	ReceiverLocationKey	Amount	Is_laundering	Is_suspicious	ach_txn_create_time	ach_txn_complete_time	txn_process_time_hours
1	1	20220111	1814127	1413990	31	60	71	71	109.01	0	0	2026-03-28 23:08:08.380	NULL	NULL
2	2	20220111	1814127	1408898	33	63	71	71	2136.08	0	0	2026-03-28 23:08:08.380	NULL	NULL
3	3	20220111	1852898	1804890	30	60	71	71	1500.52	0	0	2026-03-28 23:08:08.380	NULL	NULL
4	4	20220111	1827997	1811997	39	65	71	71	1217.07	0	0	2026-03-28 23:08:08.380	NULL	NULL
5	5	20220111	1827997	1402227	33	63	71	71	857.31	0	0	2026-03-28 23:08:08.380	NULL	NULL
6	6	20220111	1841997	1808890	33	63	71	71	7823.84	0	0	2026-03-28 23:08:08.380	NULL	NULL

Query executed successfully. DELL\MSSQLSERVER2 (17.0 RTM) DELL\UMAR (72) AML_DW 00:00:00 Row: 1, Col: 1 10 rows

TASK 6 – ACCUMULATING FACT TABLE

6.1 Introduction

This task implements an Accumulating Fact Table for transaction processing. Unlike standard fact tables, accumulating fact tables are updated over time as new events occur. In this scenario, transactions are initially recorded and later updated when they are completed.

6.2 Extending the Fact Table

The existing Fact_Transactions table was extended by adding three new columns:

- accm_txn_create_time – stores the timestamp when the record is created
- accm_txn_complete_time – stores the timestamp when the transaction is completed
- txn_process_time_hours – stores the duration between creation and completion in hours

SQL Implementation:

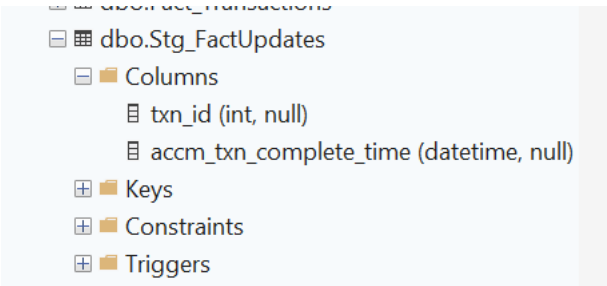
```sql
USE AML_DW;

ALTER TABLE Fact_Transactions
ADD
    accm_txn_create_time DATETIME,
    accm_txn_complete_time DATETIME,
    txn_process_time_hours INT;
```

6.3 Preparing the Update Dataset

A separate dataset was created using a staging table:

Table: Stg_FactUpdates



SQL Implementation:

```sql
CREATE TABLE Stg_FactUpdates (
    txn_id INT,
    accm_txn_complete_time DATETIME
);

INSERT INTO Stg_FactUpdates VALUES
(1, GETDATE()+1),
(2, GETDATE()+2),
(3, GETDATE()+3);

SELECT * FROM Stg_FactUpdates;
```

6.4 Results

After executing the update process:

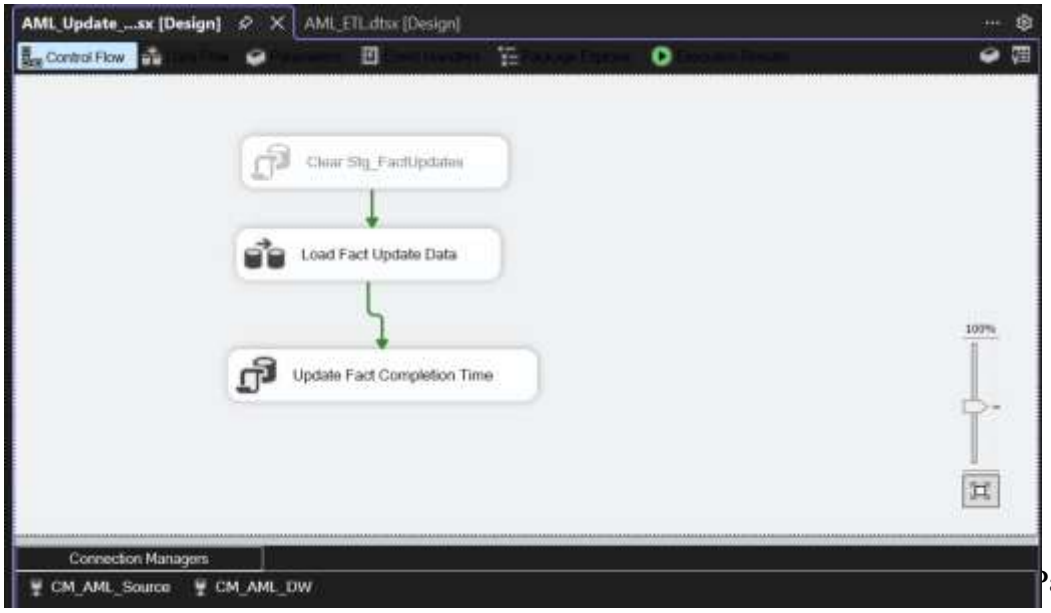
- accm_txn_complete_time was populated
- txn_process_time_hours was calculated correctly
- Fact table records were successfully updated

6.5 SSIS Package for Fact Updates

A separate SSIS package named AML_Update.dtsx was developed to update the accumulating fact table. The Control Flow consists of three main tasks:

- Clearing the staging table
- Loading update data into the staging table
- Updating the fact table with completion time and processing duration

Control Flow Design



100 % No issues found Ln: 34, Ch: 1 SPC CRL

Results Messages

	TransactionID	DateKey	SenderAccountKey	ReceiverAccountKey	PaymentTypeKey	CurrencyKey	SenderLocationKey	ReceiverLocationKey	Amount	is_laundering	is_suspicious	acct_txn_create_time	acct_txn_complete_time	txn_process_time_hours
1	1	20220111	3790771	3889538	80	154	76	76	133.31	0	0	2026-04-13 23:19:07.433	2026-04-15 02:19:07.433	27.00
2	2	20220111	3790771	3882157	82	154	76	76	2130.06	0	0	2026-04-13 23:19:07.433	2026-04-15 05:19:07.433	30.00
3	3	20220111	3868379	3821361	79	154	76	76	1500.02	0	0	2026-04-13 23:19:07.433	2026-04-15 08:19:07.433	33.00
4	4	20220111	3883434	3722201	84	154	76	76	1217.87	0	0	2026-04-13 23:19:07.433	2026-04-15 11:19:07.433	36.00
5	5	20220111	3883434	3851270	82	154	76	76	857.31	0	0	2026-04-13 23:19:07.433	2026-04-15 14:19:07.433	39.00
6	6	20220111	3885542	3842522	81	154	76	76	7859.14	0	0	2026-04-13 23:19:07.433	2026-04-15 17:19:07.433	42.00
7	7	20220111	3885542	3842522	81	154	76	76	7917.52	0	0	2026-04-13 23:19:07.433	2026-04-15 20:19:07.433	45.00
8	8	20220111	3885542	3842522	82	154	76	76	7858.09	0	0	2026-04-13 23:19:07.433	2026-04-15 23:19:07.433	48.00
9	9	20220111	3885542	3842522	82	154	76	76	7837.07	0	0	2026-04-13 23:19:07.433	2026-04-16 02:19:07.433	51.00
10	10	20220111	3885542	3842522	82	154	76	76	7832.13	0	0	2026-04-13 23:19:07.433	2026-04-16 05:19:07.433	54.00
11	11	20220111	3885542	3842522	78	154	76	76	7856.74	0	0	2026-04-13 23:19:07.433	2026-04-16 08:19:07.433	57.00
12	12	20220111	3885542	3842522	82	154	76	76	7823.64	0	0	2026-04-13 23:19:07.433	2026-04-16 11:19:07.433	60.00
13	13	20220111	3885542	3842522	79	154	76	76	7857.14	0	0	2026-04-13 23:19:07.433	2026-04-16 14:19:07.433	63.00
14	14	20220111	3885542	3842522	84	154	76	76	7915.40	0	0	2026-04-13 23:19:07.433	2026-04-16 17:19:07.433	66.00
15	15	20220111	3885542	3842522	81	154	76	76	7828.82	0	0	2026-04-13 23:19:07.433	2026-04-16 20:19:07.433	69.00
16	16	20220111	3885542	3842522	84	154	76	76	7884.21	0	0	2026-04-13 23:19:07.433	2026-04-16 23:19:07.433	72.00
17	17	20220111	3885542	3842522	84	154	76	76	7868.31	0	0	2026-04-13 23:19:07.433	2026-04-17 02:19:07.433	75.00

*# Disconnected

6.8 Conclusion

This implementation demonstrates an accumulating fact table where:

- Data is first inserted
- Then updated later as new information becomes available

This approach is useful for tracking lifecycle-based processes such as transaction completion, order processing, and workflow tracking.